

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:40

1. **Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.**

Answer:

Use Case Diagram – Book Bank System

Actors:

- Student
- Book Bank Administrator
- Central Computer
- Office Staff
- Librarian

Use Cases:

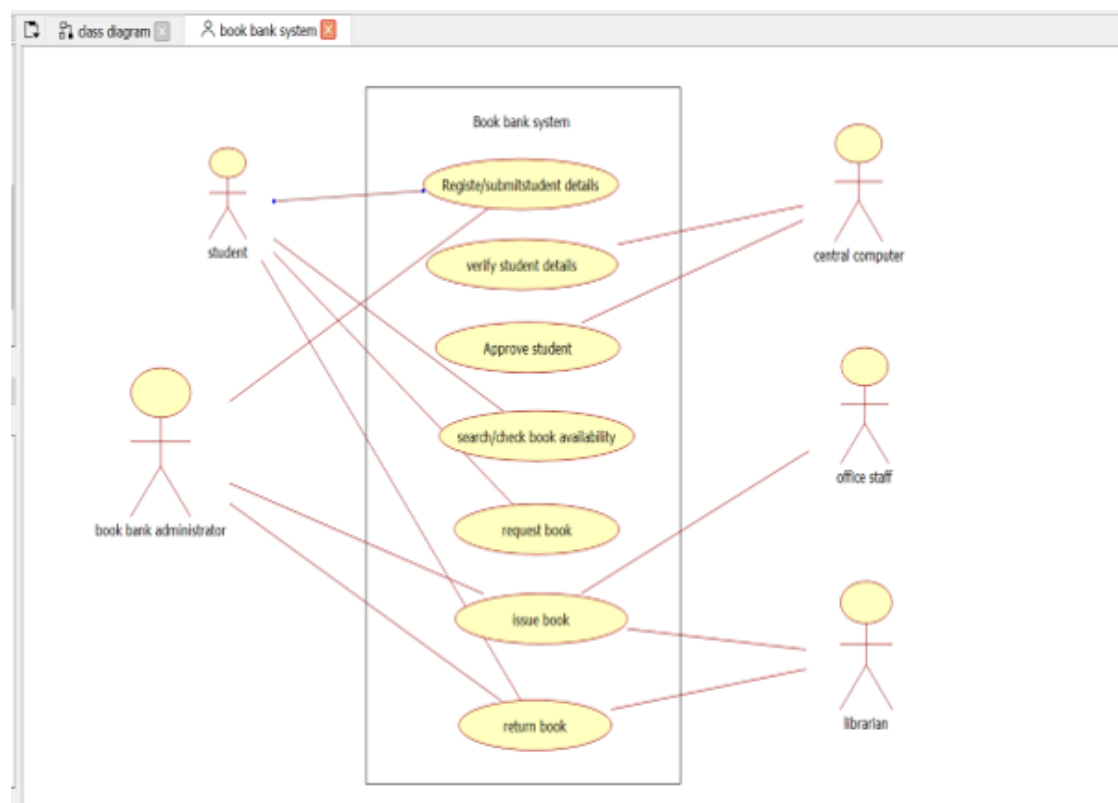
- Register/Submit Student Details
- Verify Student Details
- Approve Student
- Search/Check Book Availability
- Request Book
- Issue Book
- Return Book

Relationships/Flow:

- Student → Register/Submit Student Details

- Book Bank Administrator → Register/Submit Student Details
- Central Computer → Verify Student Details
- Central Computer → Approve Student
- Student → Search/Check Book Availability
- Student → Request Book
- Office Staff → Issue Book
- Librarian → Issue Book
- Librarian → Return Book

UML DIAGRAM :



2. **Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.**

Answer:

Class Diagram – Exam Registration System

Classes:

User

- Attributes: user_id, name, email, password
- Methods: login(), logout()

Student/Candidate

- Attributes: student_id, department, semester
- Methods: registerExam(), viewExamSchedule()

Registration

- Attributes: registration_id, registration_date, status
- Methods: registerStudent(), cancel()

Exam

- Attributes: exam_id, exam_name, date
- Methods: scheduleExam(), updateExam()

Subject

- Attributes: sub_code, sub_name
- Methods: addSub(), updateSub()

Admin

- Attributes: admin_id
- Methods: addExam(), updateExamSchedule(), manageRegistration()

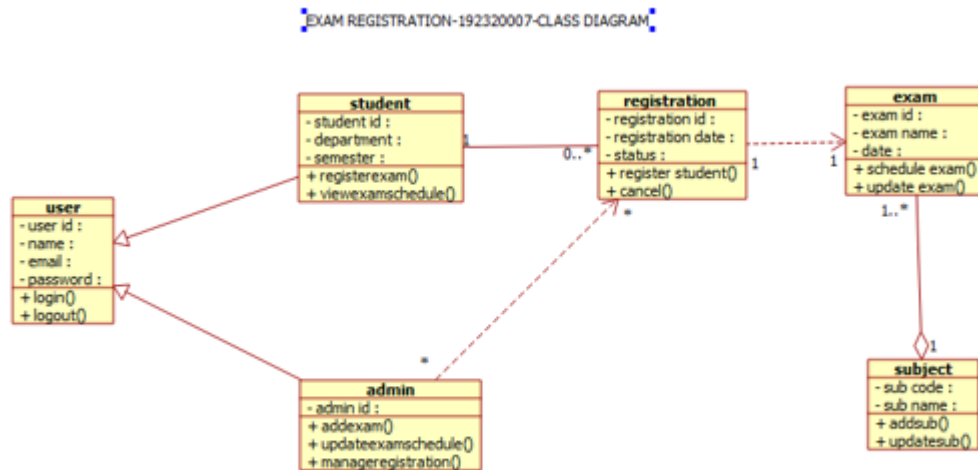
Office

- Attributes: office_id
- Methods: verifyApproval(), issueHallTicket()

Relationships:

- User → Student : **Inheritance**
- User → Admin : **Inheritance**
- Student → Registration : **1 : 0..***
- Registration → Exam : **1 : 1**
- Exam → Subject : **1 : 1..***
- Admin → Registration : **Manages**
- Office → Student : **Issues Hall Ticket**

UML DIAGRAM :



3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.

Answer:

Component Diagram – Stock Maintenance System

Components:

User Interface

- stock_interface.html

Stock Operation

- module.jar

Stock Management

- Controls stock operations and order processing.

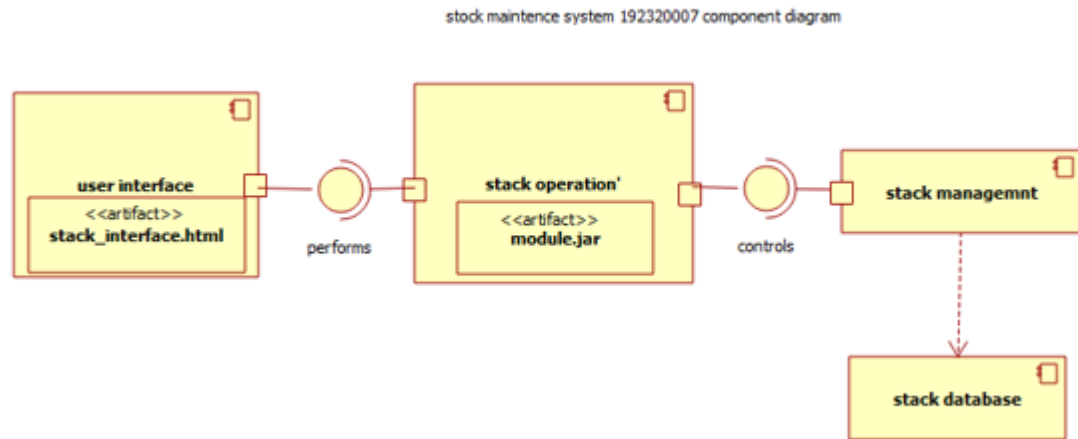
Stock Database

- Stores product, order, customer and stock details.

Connections:

- User Interface → Stock Operation — performs
- Stock Operation → Stock Management — controls
- Stock Management → Stock Database — accesses/updates

UML DIAGRAM :



Main Functions:

- Place Order
- Check Stock
- Verify Order
- Purchase Items
- Deliver Items
- Update Stock Details

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4. **Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.**

Answer:

Use Case Diagram – Online Course Reservation System

Actors:

- Student
- University/Central Management System

Use Cases:

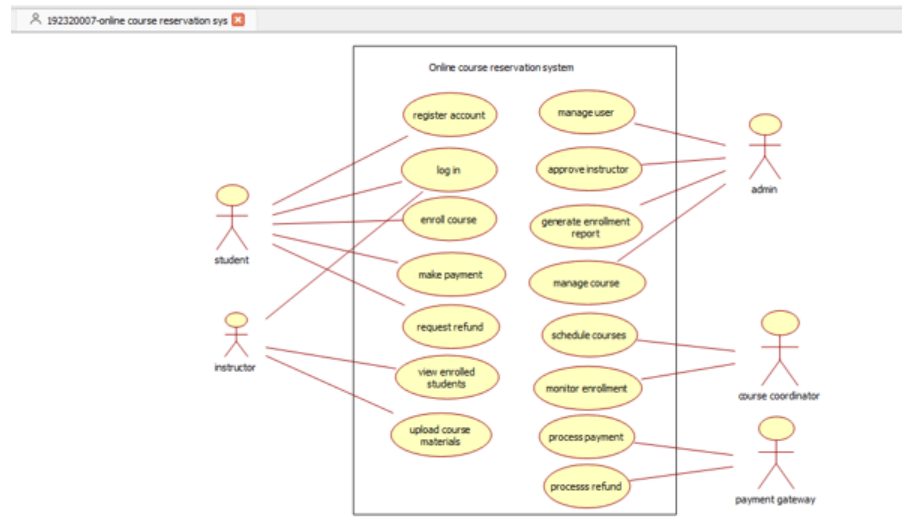
- Browse Courses
- Select Course
- Check Seat Availability
- Enroll Course
- Confirm Enrollment

Relationships/Flow:

- Student → Browse Courses

- Student → Select Course
- Central Management System → Check Seat Availability
- Central Management System → Enroll Course
- Student → Confirm Enrollment

UML DIAGRAM :



Use Case Relationship:

- Enroll Course <<include>> Check Seat Availability
- Confirm Enrollment <<include>> Enroll Course

Code of Conduct:

*I **D Varshini / 192320007** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation